1 (a) (i) **$2Fe^{3+}(aq) + Sn^{2+}(aq) \longrightarrow 2Fe^{2+}(aq) + Sn^{4+}(aq)$** [1]

ignore state symbols

(ii) $E^{\ominus}$ (Fe^{3+}/Fe^{2+}) = +0.77 V [1]
$E^{\ominus}$ (Sn^{4+}/Sn^{2+}) = +0.15 V

$E^{\ominus}_{cell} = E^{\ominus}_{reduction} - E^{\ominus}_{oxidation}$
= +0.77 – (+0.15)
= <u>+0.62 V</u>

(iii) Reduction of Fe^{3+} to Fe^{2+} by $SnCl_2$ is <u>spontaneous</u> since the $E^{\ominus}_{cell}$ calculated in **(ii)** is <u>positive</u>. [2]

$E^{\ominus}$ (Fe^{2+}/Fe) = –0.44 V
$E^{\ominus}$ (Sn^{4+}/Sn^{2+}) = +0.15 V

$E^{\ominus}_{cell}$ = –0.44 – (+0.15)
= <u>–0.59 V</u> < 0 (not spontaneous)

However, the reduction of Fe^{2+} to Fe by $SnCl_2$ is <u>not spontaneous</u> since the $E^{\ominus}_{cell}$ is <u>negative</u>.

[1]: determine $E^{\ominus}_{cell}$ for reduction of Fe^{2+}
[1]: conclusion based on both $E^{\ominus}_{cell}$

(iv) [2]

titration number	1	2	3
initial burette reading / cm^3	0.00	19.95	2.10
final burette reading / cm^3	19.95	40.05	22.15
titre / cm^3	19.95	<u>20.10</u>	<u>20.05</u>

average volume of KI used = ½(20.10 + 20.05)
= <u>20.08 cm^3</u>

$n(K_2Cr_2O_7)$ required = $\frac{20.08}{1000}$ x 0.100
= 0.002008 mol
= <u>0.00201 mol</u>

[1]: correctly determine the titre for run 2 and 3 and used it to find the average titre based on hierarchy rule (i.e. within 0.05 cm^3 in this case)
[1]: $n(K_2Cr_2O_7)$

(v) **$6Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) + 14H^{+}(aq) \longrightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$** [1]

ignore state symbols

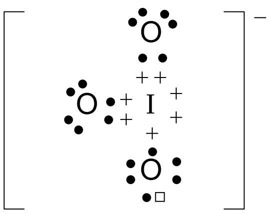
(vi) n(Fe^{2+}) in 25.0 cm^3 of solution = 6 x 0.00201 **[2]**
= 0.01206 mol
n(Fe^{2+}) in 250 cm^3 of solution = 0.01206 x 10
= <u>0.1206 mol</u>

mass of Fe present = 0.1206 x 55.8
= 6.729 g

% by mass of iron in the sample of iron ore = $\frac{6.729}{11.05}$ x 100%
= <u>60.9 %</u>

[1]: n(Fe^{2+}) originally present in 250 cm^3 (scaling)
[1]: % by mass of iron in iron ore (ecf)

(b) **(i)** [1]



□ e from metal
+ e from I
• e from O

[1]: correct dot–and–cross

(ii) [1]

$$\mathbf{Mg(IO_3)_2 \longrightarrow MgO + I_2 + \frac{5}{2} O_2}$$

(iii) [2]

- size of cations : <u>Mg^{2+}(0.065nm) > Fe^{2+}(0.061nm)</u>
- charge of the cations is the same
- <u>charge density</u> of M^{2+}: Mg^{2+} < Fe^{2+}
- <u>polarising power</u> of M^{2+}: Mg^{2+} < Fe^{2+}
- Fe^{2+} ion is able to <u>distort</u> electron cloud of IO_3^- anion to a <u>greater extent</u>. Hence, I–O covalent bond within the IO_3^- anion is <u>more weakened</u> in $Fe(IO_3)_2$.
- $Fe(IO_3)_2$ will need to be heated to a <u>lower temperature</u> for decomposition.

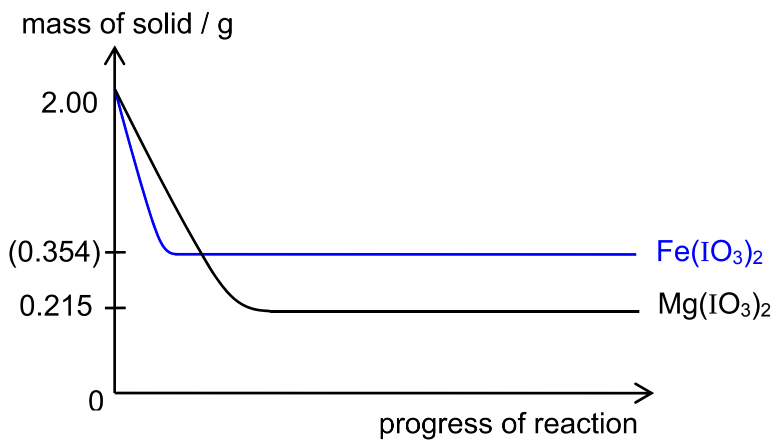
[1]: charge density / polarsising power (based on ionic radius quoted)
[1]: extent of distortion / weakening of I–O bond and lower temperature required

(accept comparable / similar temperature)

(iv) [1]

n($Mg(IO_3)_2$) reacted = $\frac{2.00}{24.3 + 2[126.9 + 3(16.0)]}$ (or $\frac{2.00}{374.1}$)
= 5.35 x 10^{-3} mol
mass of MgO left, y = 5.35 x 10^{-3} x (24.3 + 16.0)
= <u>0.215 g</u>

(v) [1]



[1]: steeper gradient and larger mass of residue

$$n(Fe(IO_3)_2) \text{ reacted} = \frac{2.00}{55.8 + 2[126.9 + 3(16.0)]} \qquad \left(\text{or } \frac{2.00}{405.6}\right)$$

$= 4.93 \times 10^{-3}$ mol

mass of FeO left, y $= 4.93 \times 10^{-3} \times (55.8 + 16.0)$

$= \underline{0.354 \text{ g}}$

2 (a) Homogenous catalyst increases the rate of a reaction where the reactant(s) and the catalyst exist in the same phase. For example, Fe^{2+}(aq) catalyses the reaction between I^-(aq) and $S_2O_8^{2-}$(aq). [2]

[1]: definition
[1]: example (accept other homogeneous system e.g. H_2O_2/Mn^{2+})

(b) (i) $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$ [1]

(ii) (homogeneous) catalyst [2]

Rate of reaction is slow without cobalt(II) but becomes vigorous in its presence. Cobalt(II) forms an intermediate (cobalt(III)) and is regenerated at the end of the reaction.

[1]: role
[1]: evidence (either one)

(iii) $E^\ominus$ (Co^{3+}/Co^{2+}) = +1.89 V [2]
$E^\ominus$ (O_2/H_2O) = +1.23 V

$E^\ominus_{cell}$ = +1.89 – (+1.23)
= +0.66 V > 0 (spontaneous)

Co^{3+}(aq) readily reduces to Co^{2+}(aq) in aqueous solution but yet Co(III) is formed in this reaction (green solution).

Tartaric acid acts as a ligand which stabilises Co(III).

[1]: shows unstability of Co^{3+}(aq)
[1]: role of tartaric acid

(c) (i) [1]

$$n(O_2) = \frac{96}{24000} = 4.00 \times 10^{-3} \text{ mol}$$

$$n(H_2O_2) = 2 \times 4.00 \times 10^{-3} = 8.00 \times 10^{-3} \text{ mol}$$

$$[H_2O_2] = \frac{8.00\times10^{-3}}{0.100} = \underline{0.0800 \text{ mol dm}^{-3}}$$

(ii) When volume of O_2(g) collected reaches 88 cm^3, 8 cm^3 of O_2 is left to be collected ($\frac{8}{96} = \frac{1}{12}$). [1]

$$1 \longrightarrow \frac{1}{2} \longrightarrow \frac{1}{4} \longrightarrow \frac{1}{8} \longrightarrow \quad \longrightarrow \frac{1}{12}\ [(\frac{1}{2})^n]$$

$\frac{C}{C_o} = (\frac{1}{2})^n$, where n = no. of half–life undergone

$$\frac{1}{12} = (\frac{1}{2})^n$$

$$n = \ln(\frac{1}{12}) \div \ln(\frac{1}{2}) = 3.58$$

estimated time taken for 88 cm^3 of O_2 collected = 3.58 x 35
= 125 min

(accept 4 x 35 = 140 min because $\frac{8}{96} \approx (\frac{1}{2})^4$ / 3.5 x 35 = 122.5 min because $\frac{1}{12}$ is between $\frac{1}{8}$ and $\frac{1}{16}$, i.e. 3rd and 4th half–life)

(iii) $k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{35}$ [2]

$= \underline{0.0198 \text{ min}^{-1}}$

(iv) time taken for H_2O_2 to decrease to the same extent (i.e. 50% in this case) will be the same regardless $[H_2O_2]$ is 0.08 mol dm^{-3} or 0.2 mol dm^{-3} because "*half–life is independent of the initial concentration*". [1]

Alternatively,
rate = k $[H_2O_2]$ (1st order reaction)
If $[H_2O_2]$ x 2.5 times, rate x 2.5 times.

Hence,

for 0.08 M H_2O_2	0.04 mol dm^{-3} decomposed in 35 min
for 0.2M H_2O_2	0.1 mol dm^{-3} decomposed in 35 min (as rate x 2.5) i.e. $\frac{0.1}{0.2}$ x 100% = 50% decomposed

(v) Titrate the $[H_2O_2]$ remaining at regular time intervals against a standard solution of $KMnO_4$. [1]
Monitor the decrease in mass of the H_2O_2 solution at regular time intervals.

[1] either one

3 (a) (i) PSI of PM10 = $\frac{200-100}{350-150}(320-150)+100$ = $\underline{185}$ [2]

PSI of CO = $\frac{300-200}{34-17}(20-17)+200$ = $\underline{218}$

[1] each

(ii) overall PSI is the maximum value out of 185, 218, 112, 133 and 150. [1]
Hence overall PSI is 218. ecf

(iii) I would advise the PE teacher to avoid strenuous physical exertion / conduct the lesson indoor (words to the effect based on valid reasoning). [1] ecf

(b) (i) $SO_2 + 2H_2O \longrightarrow SO_4^{2-} + 4H^+ + 2e^-$ (x 5) [1]
$2IO_3^- + 12H^+ + 10e^- \longrightarrow I_2 + 6H_2O$

overall: $\mathbf{5SO_2 + 4H_2O + 2IO_3^- \longrightarrow 5SO_4^{2-} + 8H^+ + I_2}$

(ii) $H^+ + OH^- \longrightarrow H_2O$ [2]

n(NaOH) reacted = 0.01 x 0.005
= 5 x 10^{-5} mol
n(H^+) reacted with NaOH = 5 x 10^{-5}

n(SO_2) in 1 m^3 sample of air = 5 x 10^{-5} x $\frac{5}{8}$
= $\underline{3.125 \times 10^{-5} \text{ mol}}$
mass of SO_2 in 1 m^3 sample of air = 3.125 x 10^{-5} x [32.1 + 2(16.0)] (or 64.1)
= 0.00200 g
= 2000 μg
concentration of SO_2 = $\underline{2000}$ μg m^{-3}

[1]: [SO_2] in mol m^{-3}
[1]: [SO_2] in μg m^{-3}

(c) (i) In 1 m^3, [1]
mass of air is 1 kg
mass of PM10 is $\frac{2\times10^{-5}}{100}$ x 1 = 2 x 10^{-7} kg = 0.0002 g = 200 μg
Hence concentration of PM10 is $\underline{200}$ μg m^{-3}

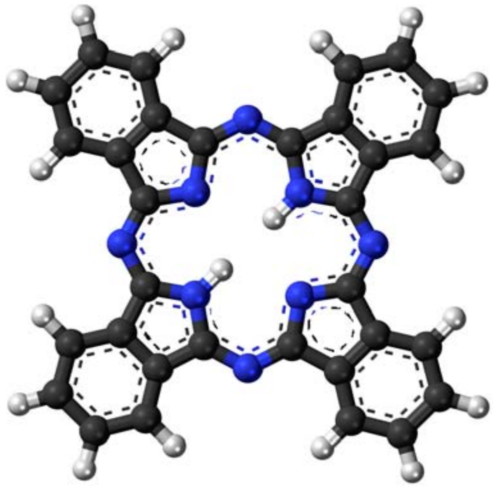
(ii) Since concentration calculated in **(c)(i)** is more than 180, the sample of air has exceeded the limit. [1] ecf

The German city of Leipzig will be fined.

(d) (i) $\mathbf{C_{32}H_{18}N_8}$ [1]

(ii) It contains a ring of delocalised pi electron clouds / the p orbitals of the N and C atoms can overlap sideways to form conjugated ring system of pi bonds. [1]

FYI



ball and stick diagram

(e) **(i)** copper and N1: ionic [2]
copper and N2: dative

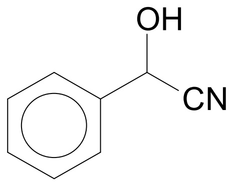
[1] each

(ii) electronic configuration of Cu is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ [1]
electronic configuration of Cu^{2+} is $\mathbf{1s^2 2s^2 2p^6 3s^2 3p^6 3d^9}$

(iii) **8** [1]

(iv) presence of C*l* atoms alters the extent of d–orbital splitting by the ligands and hence energy absorbed by electron (d–d transition) and colour observed will be different [1]

4 **(a)** **(i)** [3]



step 1: HCN in the presence of NaCN (or trace of NaOH)

step 2: $CH_3CH_2CH_2CH_2CH_2COCl$ (do not accept its carboxylic acid)

[1]: structure
[1] each reagent

(ii) The instananeous dipole–induced dipole attractions between the non–polar benzene ring / pentyl ($–(CH_2)_4CH_3$) group in mandelonitrile hexanoate molecule and the non–polar hexane molecule are comparable in strength to that between mandelonitrile hexanoate molecules and between hexane molecules. [1]

(b) **(i)** Hydrolysis (or nucleophilic acyl substitution) of ester and oxidation of 2° alcohol [2]

[1] each

(ii) the C–C bond in COCN is stronger than the C–C*l* bond in COC*l* [1]
or CN^- in COCN is poorer leaving group than Cl^- in COC*l*
or the carbonyl C in COCN is less electron deficient than in COC*l*

[1] either one

(iii) Add Tollens' reagent and warm, if a silver mirror (or grey ppt) is formed, benzaldehyde is present. Hence, pathway II has occurred. Otherwise, pathway I has occurred. [2]

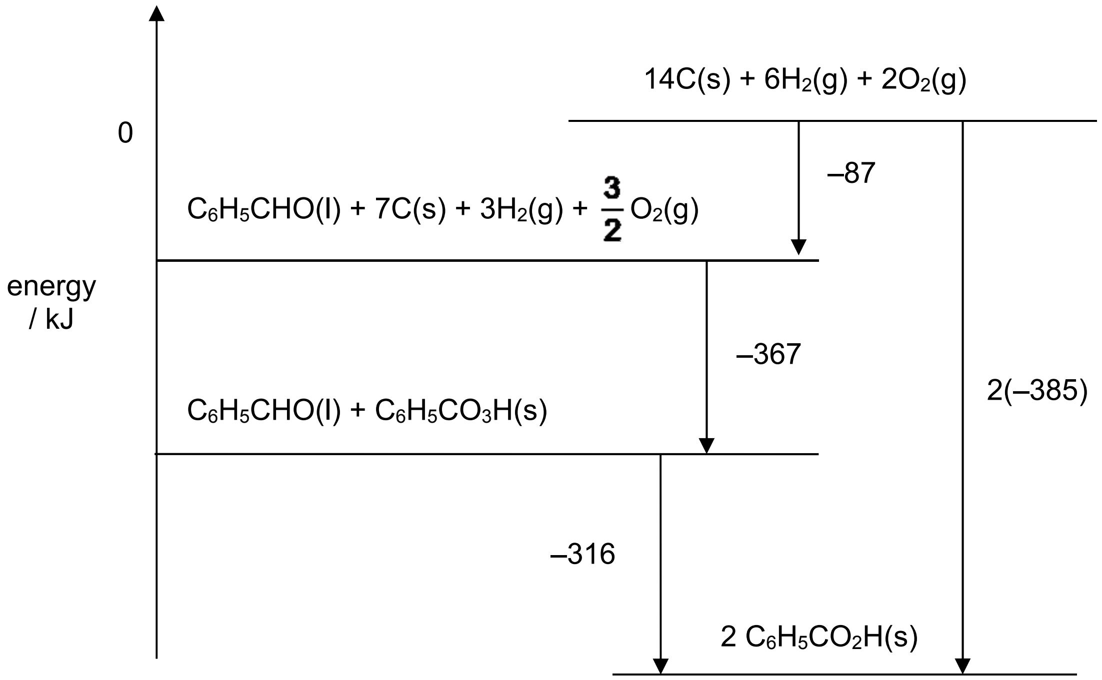
(accept 2,4–dinitrophenylhydrazine, $Cr_2O_7^{2-}/H^+$, MnO_4^-/H^+ / yellow (orange) ppt, orange solution turned green, purple solution decolourised due to benzaldehyde)

(c) **(i)** The amount of heat absorbed or evolved when one mole of substance is formed from its constituent elements in their standard states *at* 298 K and 1 bar. [1]

(ii) $-316 = 2\Delta H_f^{\ominus}(C_6H_5CO_2H) - [(-367) + (-87)]$ [1]

$\Delta H_f^{\ominus}(C_6H_5CO_2H) = \underline{-385 \text{ kJ mol}^{-1}}$

(iii) [3]



[1]: constituent elements at the highest energy level (assigned 0)
[1]: corresponding compounds at the 3 other energy levels
[1]: correct stoichiometry, state symbols and arrow directions

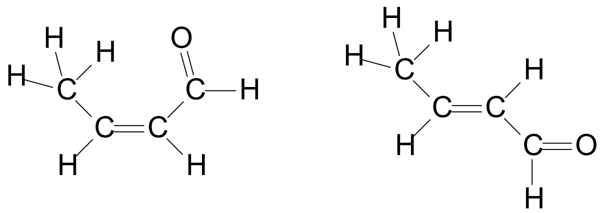
5 (a) (i) stage I: $K_2Cr_2O_7$, dilute H_2SO_4, warm with immediate distillation [2]

stage III: H_2, Pt / Pd, room temperature (accept Ni, with or w/o warm/heat)

[1] each

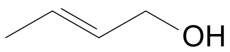
(ii) $$C_4H_{10}O + \frac{11}{2}O_2 \longrightarrow 3CO_2 + CO + 5H_2O$$ [1]

(iii) cis–trans isomerism [2]



cis **trans**

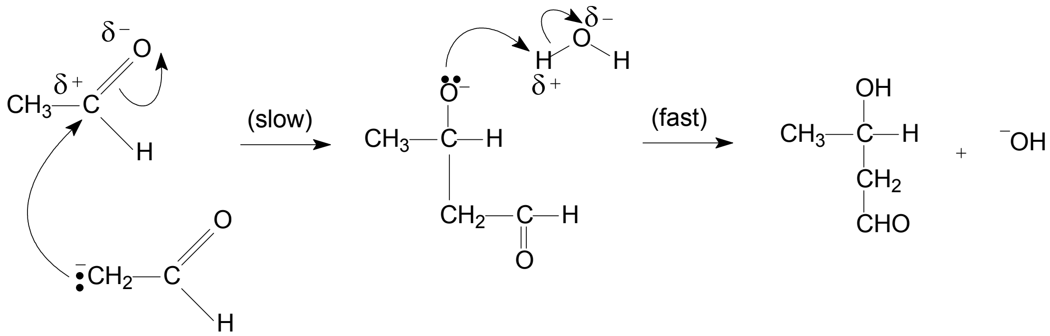
[1]: type of isomerism with correct label
[1]: displayed structure

(iv)  [2]

G is reduced to **H** (an alcohol) which reacts with sodium metal to give effervescence of H_2. **H** does not react with 2,4–DNPH due to absence of aldehyde/ketone. It reacts with aqueous Br_2 because of the C=C bond present.

[1]: structure of **H**
[1]: explanation (alcohol and C=C essential; do not accept hydroxyl or –OH group in place of alcohol)

(b) (i) [2]

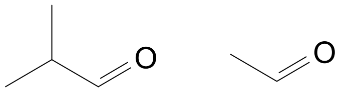


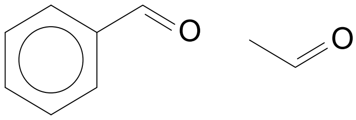
[1]: dipoles (BOD if missing on H_2O), charges and lone pairs
[1]: curly arrows and intermediate

(ii) Able to lose H and OH on adjacent C atoms in 2 different ways, giving rise a different product formed. [1]
Absence of resonance stabilised / delocalised structure / less substituted alkene compared to **G**, resulting in lower yield.

The other product is $CH_2{=}CHCH_2CHO$.

(iii) [3]

I 

II 

[1]: ethanal (CH_3CHO) for both
[1] each for the other 2

(c) **(i)** [2]

$CH_3CH_2CH(OH)CH_3$	$CH_3CH(CH_3)CH_2OH$	$(CH_3)_3COH$
J	**K**	**L**
2°	1°	3°

[1]: all 3 structural isomers correctly identified and labelled (in any order)
[1]: corresponding classification of 3 alcohols

(ii) **J** reacts with alkaline aqueous iodine [2]

$$CH_3CH_2CH(OH)CH_3 + 4I_2 + 6OH^- \longrightarrow CH_3CH_2CO_2^- + CHI_3 + 5I^- + 5H_2O$$

[1]: correctly identified the alcohol with positive reaction with I_2/OH^-
[1]: balanced equation